

OOAD PRESENTATION

Detailed Design

REFINING ARCHITECTURE INTO DIAGRAMS

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SUBJECT

Analysis & Design



Objectives



Visualizing Structure

Bridge the gap between abstract requirements and concrete implementation details.



Refining Logic

Define exact attributes, operations, and strict data types for coding.



Mastering UML

Learn to craft industry-standard Class and Package diagrams.



What is Detailed Design?

Moving from "What" to "How".

It acts as the blueprint for programmers. It specifies the internal logic, data structures, and interface definitions.

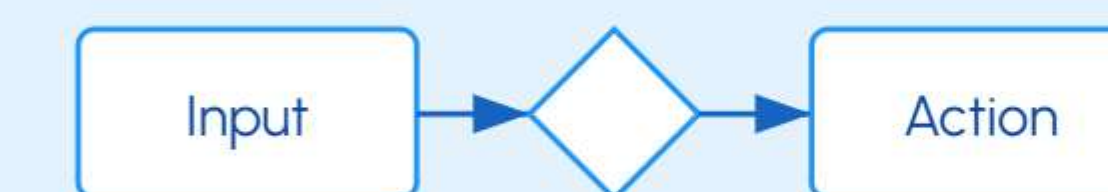
- ✓ Algorithm Logic
- ✓ Data Type Selection
- ✓ Interface Signatures

Detailed Design Specification

// 1. Interface Signatures

```
public Response processOrder(Order input);
```

// 2. Algorithm Logic



// 3. Data Structure Selection

```
Map<Integer, Item> inventory = new HashMap<>();
```

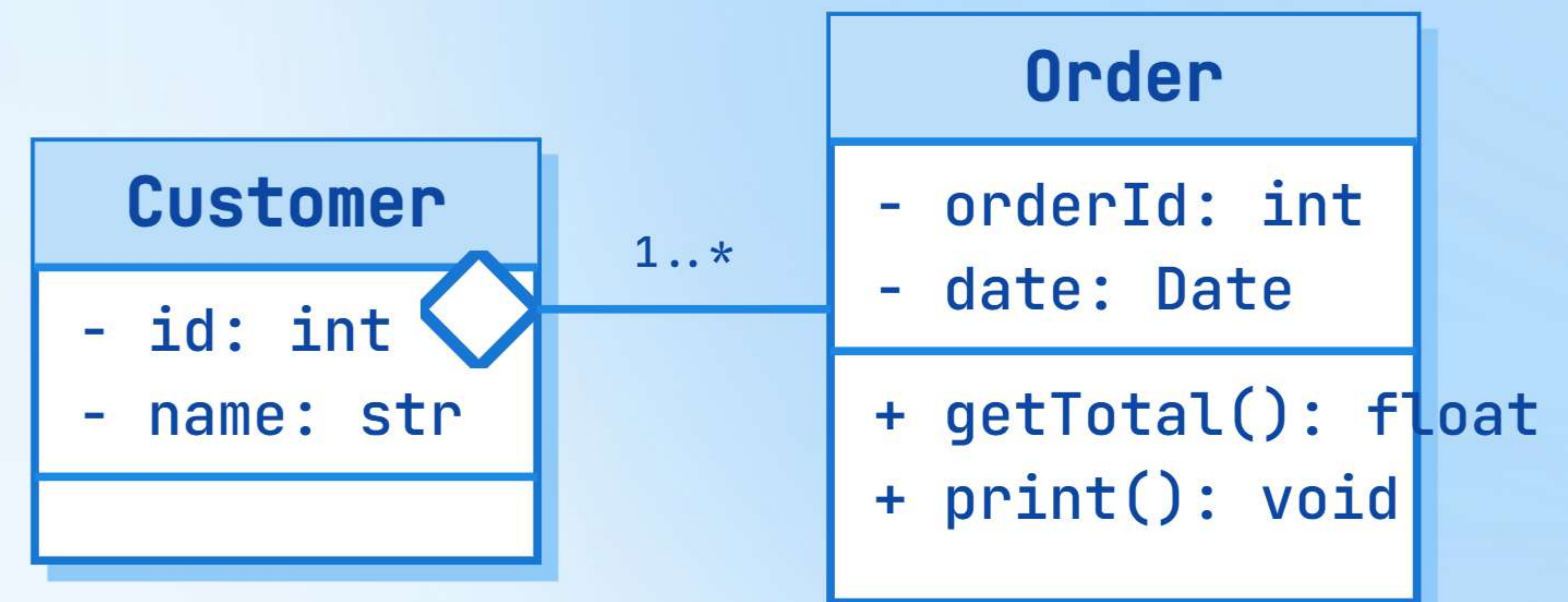


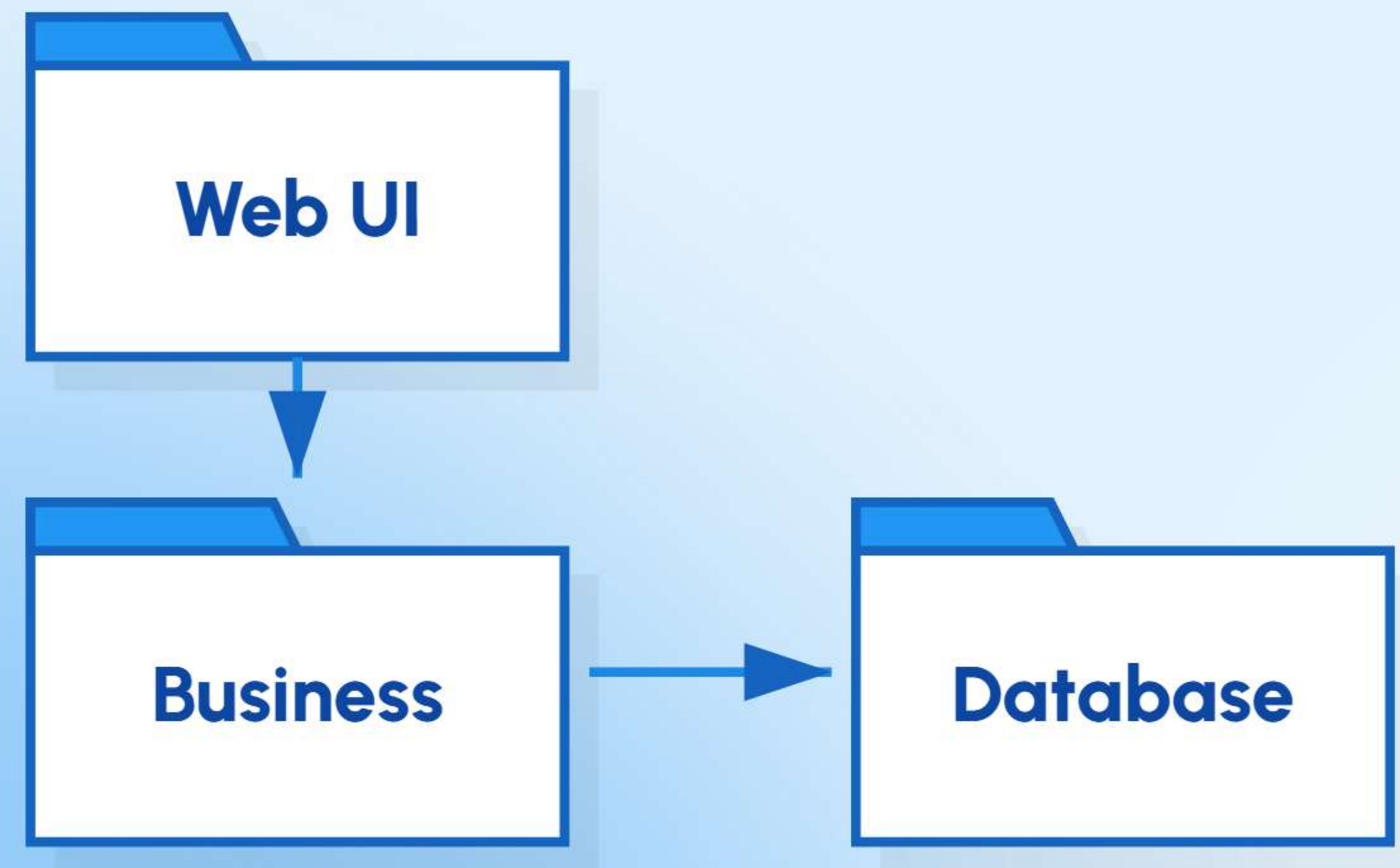
Class Diagrams

The static structure of the system. It shows classes, their attributes, operations, and the relationships.

Components:

1. Class Name
2. Attributes (- private)
3. Operations (+ public)





Package Diagrams

Organizing elements into groups. Essential for large system architecture.

Namespace: Ensures class names don't collide.

Visibility: Controls what parts of the system can see others.

Deployment: Maps directly to JARs, DLLs, or folders.



The Refinement Process

1

Identify

Extract nouns from requirements.

2

Relate

Define Is-A and Has-A links.

3

Detail

Add types and visibility.

4

Group

Organize into logical packages.

Requirements → High Level Design → Detailed Design → Implementation

